Motion Deblurring of License Plate Images

Computer Vision Project

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Big Data & AI

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1 Introduction

Motion blur is a common issue in real-world imaging systems, particularly in traffic surveillance where vehicles move at high speed. This degradation significantly reduces the readability of license plates, affecting applications such as automatic number plate recognition (ANPR).

The objective of this project is to restore motion-blurred license plate images using classical image processing techniques. We compare two main approaches: inverse filtering and Wiener filtering, and evaluate their performance both visually and quantitatively.

2 Image Degradation Model

A degraded image can be modeled as:

$$g = f * h + n \tag{1}$$

where:

- f is the original image
- h is the blur kernel (motion blur)
- n is additive noise
- $*$ denotes convolution

In this project, motion blur is simulated using a linear kernel defined by a given length and angle.

3 Methodology

3.1 Dataset

The dataset consists of cropped license plate images converted to grayscale for processing.

3.2 Blur Simulation

Motion blur is artificially applied to simulate real-world degradation. Parameters include:

- Blur length
- Blur angle

3.3 Deblurring Methods

3.3.1 Inverse Filtering

Inverse filtering attempts to directly invert the degradation function in the frequency domain. However, it is highly sensitive to noise and often leads to unstable results.

3.3.2 Wiener Filtering

Wiener filtering introduces a regularization term to balance between restoration and noise suppression. It is more robust in practical scenarios.

3.4 Evaluation Metrics

Two quantitative metrics are used:

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)

3.5 Enhancement Pipeline

After restoration, a post-processing pipeline is applied:

- CLAHE (contrast enhancement)
- Sharpening (unsharp masking)
- Adaptive thresholding

4 Results

4.1 Quantitative Results

Method	PSNR	SSIM
Blurred Image	24.65	0.739
Inverse Filter	9.69	0.121
Wiener Filter	22.21	0.730

Table 1: Comparison of restoration methods

4.2 Visual Results

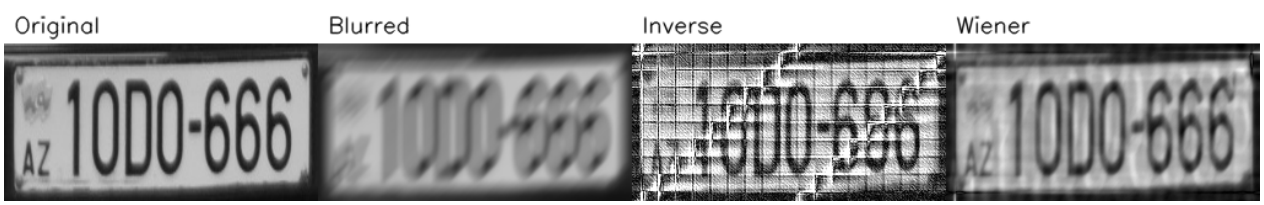

Figure 1: Original vs Blurred vs Inverse vs Wiener

5 Enhancement Results

Figure 2: Post-processing results

6 Discussion

The results clearly show that inverse filtering is unstable and amplifies noise, leading to poor image quality. In contrast, Wiener filtering provides a better balance between deblurring and noise suppression.

Although Wiener filtering does not fully restore the original image, it preserves structural information and significantly improves visual quality.

Post-processing techniques further enhance the readability of license plate characters.

7 Limitations

- Synthetic blur does not fully represent real-world conditions
- Wiener filter requires parameter tuning
- No automatic plate detection or OCR included

8 Conclusion

This project demonstrates that Wiener deconvolution is a robust approach for restoring motion-blurred license plate images. It outperforms inverse filtering both visually and quantitatively.

The addition of enhancement techniques further improves the readability of the results, making the approach suitable for practical computer vision applications.

9 References

- Gonzalez, R. C., & Woods, R. E. *Digital Image Processing*
- OpenCV Documentation
- Wiener Filtering Theory